

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 60%

QUESTIONS: 6

Total Marks:60

Annexure A

Application Based Questions

Code of Conduct:

I B. Harshitha (192420006) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

B. Harshitha

Annexure A

Application Based Questions

1. Use Case Diagram Analysis

A **Smart Hotel Reservation System** allows guests to search rooms, make reservations, cancel bookings, and process payments. Managers handle room allocation and reports, while payment services process transactions. The current design does not clearly distinguish actor responsibilities.

Analyze the system requirements, identify missing actors and use cases, and design an optimized Use Case Diagram showing proper relationships (include, extend, generalization) for better system clarity.

(10 Marks)

2. Class Diagram Design

A **University Management System** includes entities such as Student, Faculty, Course, Department, and Examination. The existing design lacks proper attributes, operations, and relationships among classes.

Analyze the system, identify relevant classes with attributes and methods, define relationships (association, aggregation, inheritance), and construct a well-structured Class Diagram.

(10 Marks)

3. Sequence Diagram Evaluation

In a **Vehicle Rental System**, a customer searches for a vehicle, books it, makes payment, and receives a rental confirmation. The interaction among system components is incomplete and poorly documented.

Analyze the process flow, identify missing objects and message sequences, and design a detailed Sequence Diagram representing proper interaction among system components.

(10 Marks)

4. State Diagram Construction

A **Traffic Signal Control System** operates through states such as Red Signal, Green Signal, Yellow Signal, Emergency Mode, and Maintenance Mode. The transitions between states are not clearly defined.

Analyze the system behavior, identify valid states and transitions, and construct a State Transition Diagram ensuring all events and conditions are properly represented.

(10 Marks)

5. Component Diagram Design

A **Smart Healthcare Portal** consists of modules such as Patient Management, Appointment Scheduling, Medical Records, Billing, and Notification Services. The current architecture does not clearly define component interactions and dependencies.

Analyze the system architecture, identify major components and their interactions, and design a Component Diagram showing dependencies and interfaces between modules.

(10 Marks)

6. Deployment Diagram Design

A **Smart City Waste Management System** is deployed across IoT waste-bin sensors, edge devices, cloud servers, and monitoring dashboards. The current design lacks clarity regarding physical architecture and communication links.

Analyze the deployment environment, identify nodes, artifacts, and communication links, and construct a detailed Deployment Diagram representing the complete system architecture.

(10 Marks)

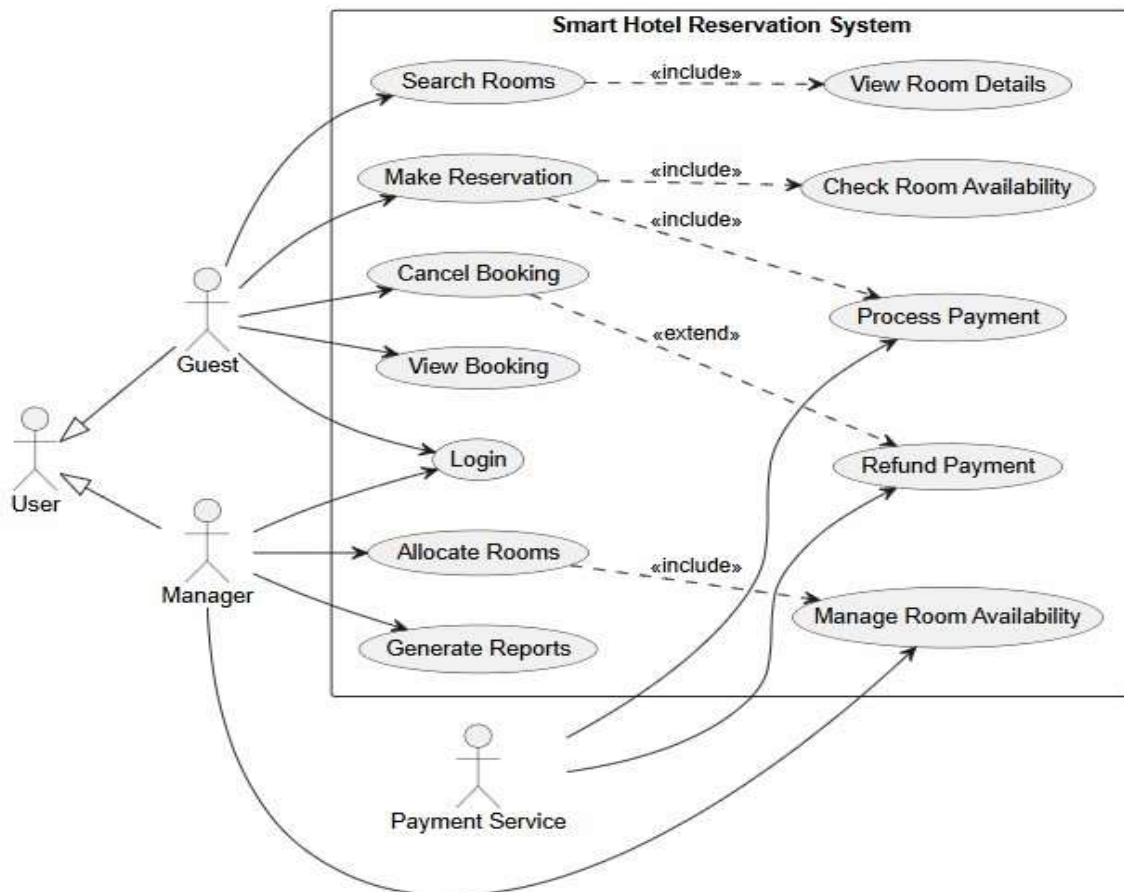
RUBRICS FOR CALCULATION

Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

1. Use Case Diagram Analysis

A **Smart Hotel Reservation System** allows guests to search rooms, make reservations, cancel bookings, and process payments. Managers handle room allocation and reports, while payment services process transactions. The current design does not clearly distinguish actor responsibilities.



System Analysis :

The Smart Hotel Reservation System enables guests to search for available rooms, make reservations, cancel bookings, and complete online payments. Managers are responsible for allocating rooms, managing room availability, and generating reports. An external payment service securely handles payment transactions and refunds. The optimized use case diagram clearly separates the responsibilities of each actor and improves system understanding.

a) Actors Identified

- User (General Actor)
- Guest
- Manager
- Payment Service (External Actor)

b) Primary Use Cases

- Login
- Search Rooms
- View Room Details
- Make Reservation
- Check Room Availability
- Process Payment
- Cancel Booking
- Refund Payment
- View Booking
- Allocate Rooms
- Manage Room Availability
- Generate Reports

c) Relationships

Generalization

- Guest and Manager inherit from the User actor.

<<include>>

- Make Reservation includes Check Room Availability.
- Make Reservation includes Process Payment.
- Search Rooms includes View Room Details.
- Allocate Rooms includes Manage Room Availability.

<<extend>>

- Cancel Booking extends Refund Payment whenever a refund is applicable.

d) Missing Actors and Use Cases

Missing Actor

- Payment Service is added as an external actor to securely process payments and refunds.

Missing Use Cases

- Login
- View Room Details
- View Booking
- Manage Room Availability
- Refund Payment

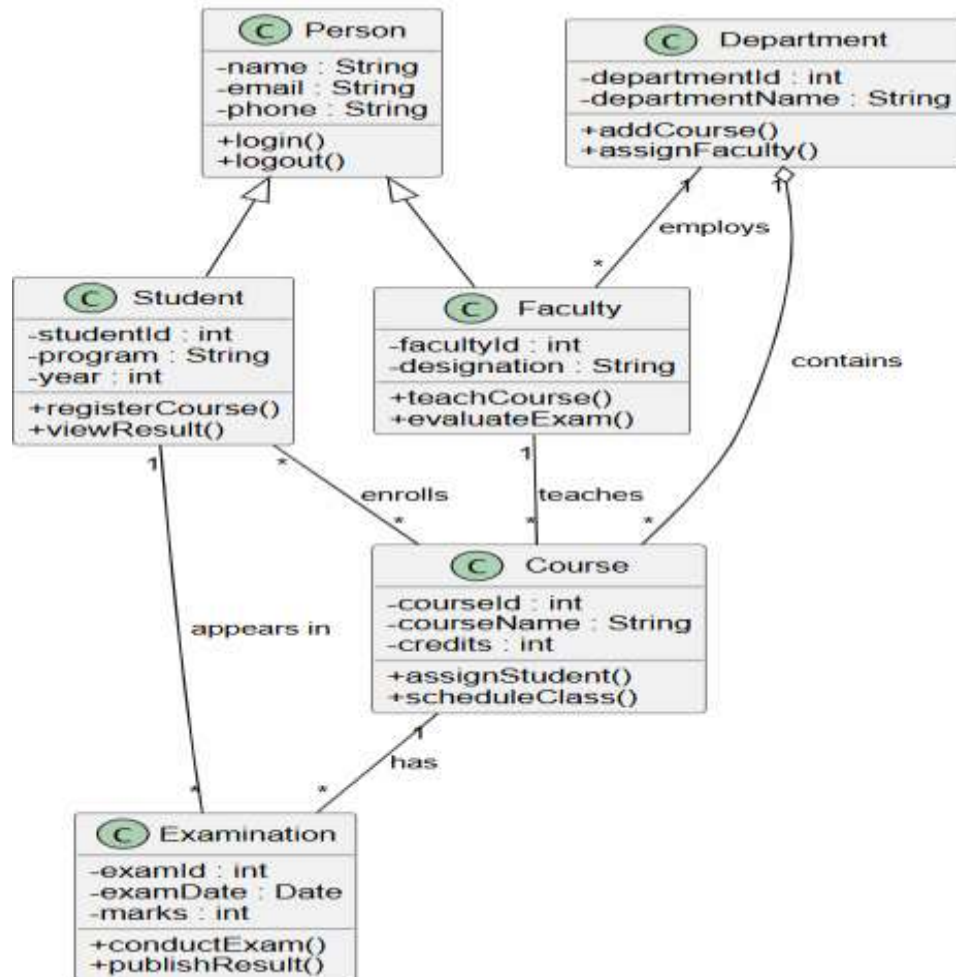
These use cases improve the completeness of the hotel reservation system.

Conclusion

The optimized Use Case Diagram clearly defines actor responsibilities and system functionalities using generalization, include, and extend relationships. It provides a complete representation of the Smart Hotel Reservation System and improves clarity and maintainability.

2. Class Diagram Design

A **University Management System** includes entities such as Student, Faculty, Course, Department, and Examination. The existing design lacks proper attributes, operations, and relationships among classes.



System Analysis :

The University Management System manages students, faculty, departments, courses, and examinations. Students enroll in courses and appear for examinations. Faculty members teach courses and evaluate examinations, while departments manage both courses and faculty. The Class Diagram defines the structure of the system by specifying attributes, operations, relationships, and multiplicity.

a) Classes with Attributes and Methods

i) Person

Attributes

- name
- email
- phone

Methods

- login()
- logout()

ii) Student

Attributes

- studentId
- program
- year

Methods

- registerCourse()
- viewResult()

iii) Faculty

Attributes

- facultyId
- designation

Methods

- teachCourse()
- evaluateExam()

iv) Department

Attributes

- departmentId
- departmentName

Methods

- addCourse()
- assignFaculty()

v) Course

Attributes

- courseId
- courseName
- credits

Methods

- assignStudent()
- scheduleClass()

vi) Examination

Attributes

- examId
- examDate
- marks

Methods

- conductExam()
- publishResult()

b) Relationships

Inheritance

- Student and Faculty inherit from the Person class.

Association

- Students enroll in Courses.
- Faculty teach Courses.
- Students appear for Examinations.
- Courses have Examinations.

Aggregation

- A Department contains multiple Courses.
- A Department employs multiple Faculty members.

c) Multiplicity

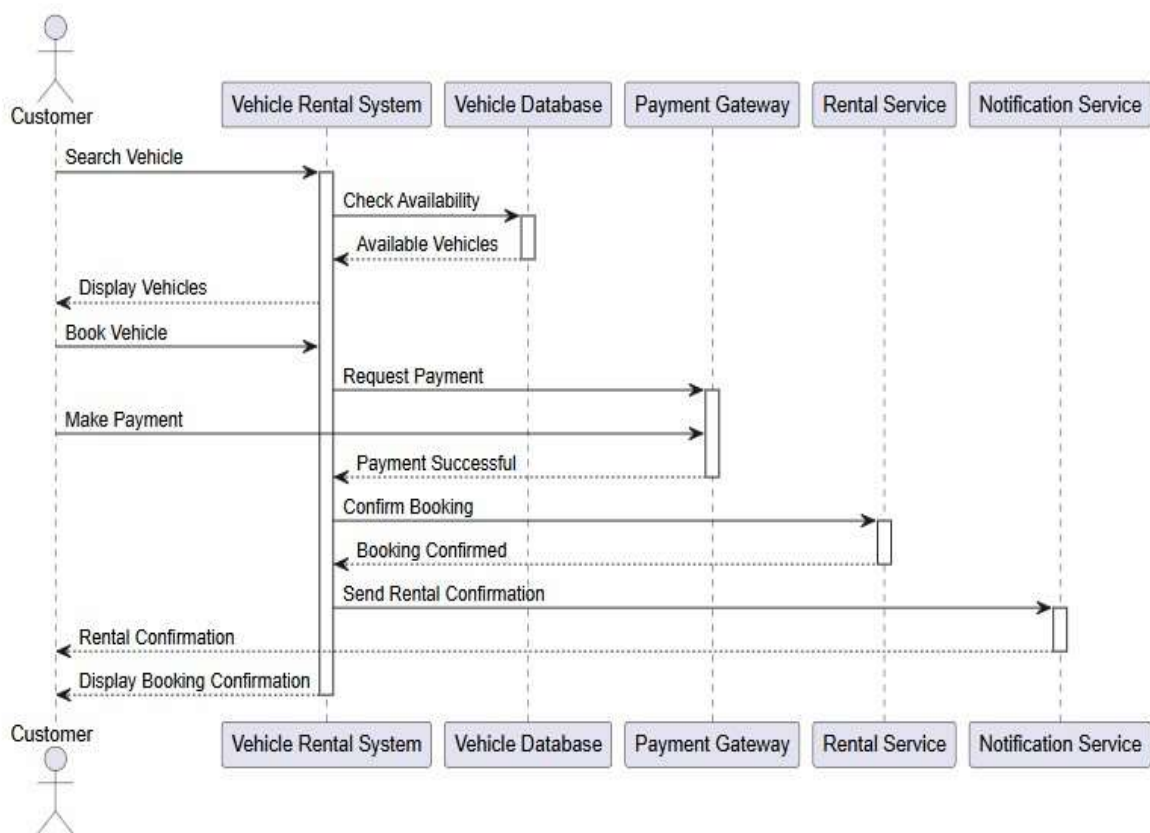
- One Department contains many Courses (1..*).
- One Department employs many Faculty members (1..*).
- One Faculty teaches many Courses (1..*).
- Many Students can enroll in many Courses (*..*).
- One Course has many Examinations (1..*).
- One Student can appear in many Examinations (1..*).

d) Conclusion

The optimized Class Diagram represents the static structure of the University Management System by defining classes, attributes, methods, inheritance, associations, aggregations, and multiplicity. It provides a clear and organized model for managing academic activities.

3. Sequence Diagram Evaluation

In a **Vehicle Rental System**, a customer searches for a vehicle, books it, makes payment, and receives a rental confirmation. The interaction among system components is incomplete and poorly documented.



System Analysis :

The Vehicle Rental System allows customers to search available vehicles, make bookings, complete online payments, and receive rental confirmations. The system interacts with the vehicle database, payment gateway, rental service, and notification service to ensure a smooth rental process. The Sequence Diagram illustrates the interaction among these components in the correct order.

a) Objects Identified

- Customer
- Vehicle Rental System
- Vehicle Database
- Payment Gateway
- Rental Service
- Notification Service

b) Message Sequence

1. Customer searches for a vehicle.
2. The system checks vehicle availability in the database.
3. The database returns the list of available vehicles.
4. The system displays the available vehicles.
5. The customer selects and books a vehicle.
6. The system requests payment through the payment gateway.
7. The customer completes the payment.
8. The payment gateway confirms the successful transaction.
9. The system sends the booking details to the rental service.
10. The rental service confirms the booking.
11. The system sends a rental confirmation through the notification service.
12. The customer receives the rental confirmation.

c) Missing Objects and Message Sequences

Missing Objects

- Vehicle Database
- Payment Gateway
- Rental Service
- Notification Service

Missing Message Sequences

- Check vehicle availability
- Payment request and confirmation
- Booking confirmation from rental service
- Rental confirmation notification to the customer

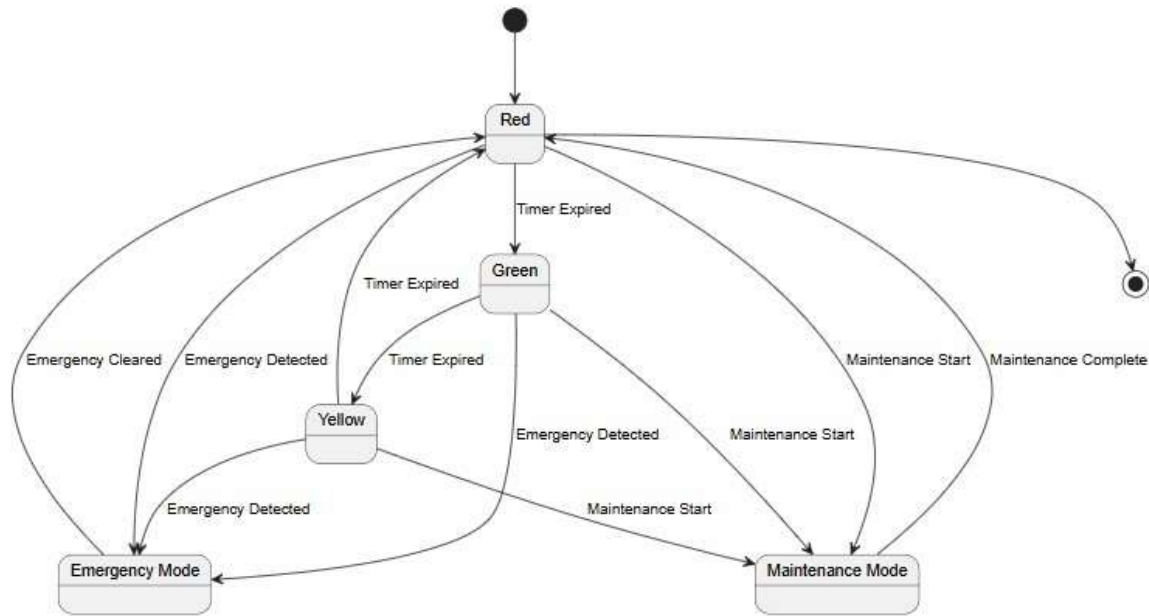
These additions make the interaction complete and well-structured.

d) Conclusion

The optimized Sequence Diagram clearly represents the interaction among all system components. It includes object lifelines, activation bars, and message sequences, providing a complete workflow from vehicle search to rental confirmation.

4. State Diagram Construction

A **Traffic Signal Control System** operates through states such as Red Signal, Green Signal, Yellow Signal, Emergency Mode, and Maintenance Mode. The transitions between states are not clearly defined.



System Analysis :

The Traffic Signal Control System controls traffic by cycling through Red, Green, and Yellow signal states. It also supports Emergency Mode to prioritize emergency vehicles and Maintenance Mode for system servicing. The State Transition Diagram illustrates the valid states, events, and transitions that ensure safe and efficient traffic signal operation.

a) Valid States

- Initial State
- Red Signal
- Green Signal
- Yellow Signal
- Emergency Mode
- Maintenance Mode
- Final State

b) Valid Transitions

- Red → Green when the timer expires.
- Green → Yellow when the timer expires.
- Yellow → Red when the timer expires.
- Red, Green, or Yellow → Emergency Mode when an emergency vehicle is detected.
- Emergency Mode → Red after the emergency is cleared.
- Red, Green, or Yellow → Maintenance Mode when maintenance begins.
- Maintenance Mode → Red after maintenance is completed.

c) Events and Conditions

Events

- Timer Expired
- Emergency Detected
- Emergency Cleared
- Maintenance Start
- Maintenance Complete

Conditions

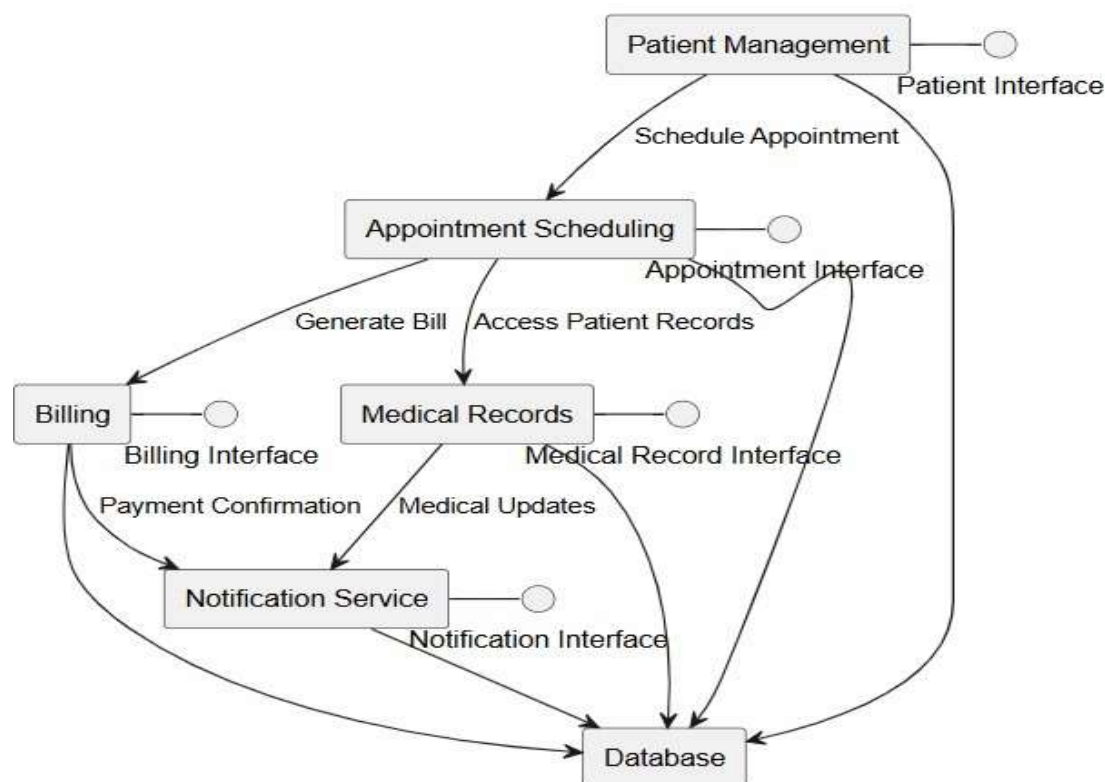
- Normal operation follows the sequence Red → Green → Yellow → Red.
- Emergency Mode interrupts the normal cycle immediately.
- Maintenance Mode can be entered from any normal signal state.
- After emergency handling or maintenance, the system returns to the Red Signal state before resuming normal operation.

d) Conclusion

The State Transition Diagram clearly models the behavior of the Traffic Signal Control System by representing all valid states, transitions, events, and conditions. It ensures safe signal operation while supporting emergency handling and maintenance activities.

5. Component Diagram Design

A **Smart Healthcare Portal** consists of modules such as Patient Management, Appointment Scheduling, Medical Records, Billing, and Notification Services. The current architecture does not clearly define component interactions and dependencies.



System Analysis :

The Smart Healthcare Portal is composed of independent components that work together to provide healthcare services. The portal manages patient information, appointment scheduling, medical records, billing, and notifications. A centralized database stores and shares information among all components. The Component Diagram illustrates the dependencies and interfaces between these modules.

a) Major Components

- Patient Management
- Appointment Scheduling
- Medical Records
- Billing
- Notification Service
- Database

b) Component Interactions

- Patient Management sends appointment requests to the Appointment Scheduling module.
- Appointment Scheduling accesses Medical Records when scheduling appointments.
- Appointment Scheduling forwards billing information to the Billing module.
- Billing sends payment confirmation to the Notification Service.
- Medical Records send medical updates to the Notification Service.
- All components interact with the Database to store and retrieve information.

c) Dependencies and Interfaces

Dependencies

- Patient Management → Appointment Scheduling
- Appointment Scheduling → Medical Records
- Appointment Scheduling → Billing
- Billing → Notification Service
- Medical Records → Notification Service
- All components depend on the Database.

Interfaces

- Patient Interface
- Appointment Interface
- Medical Record Interface
- Billing Interface
- Notification Interface

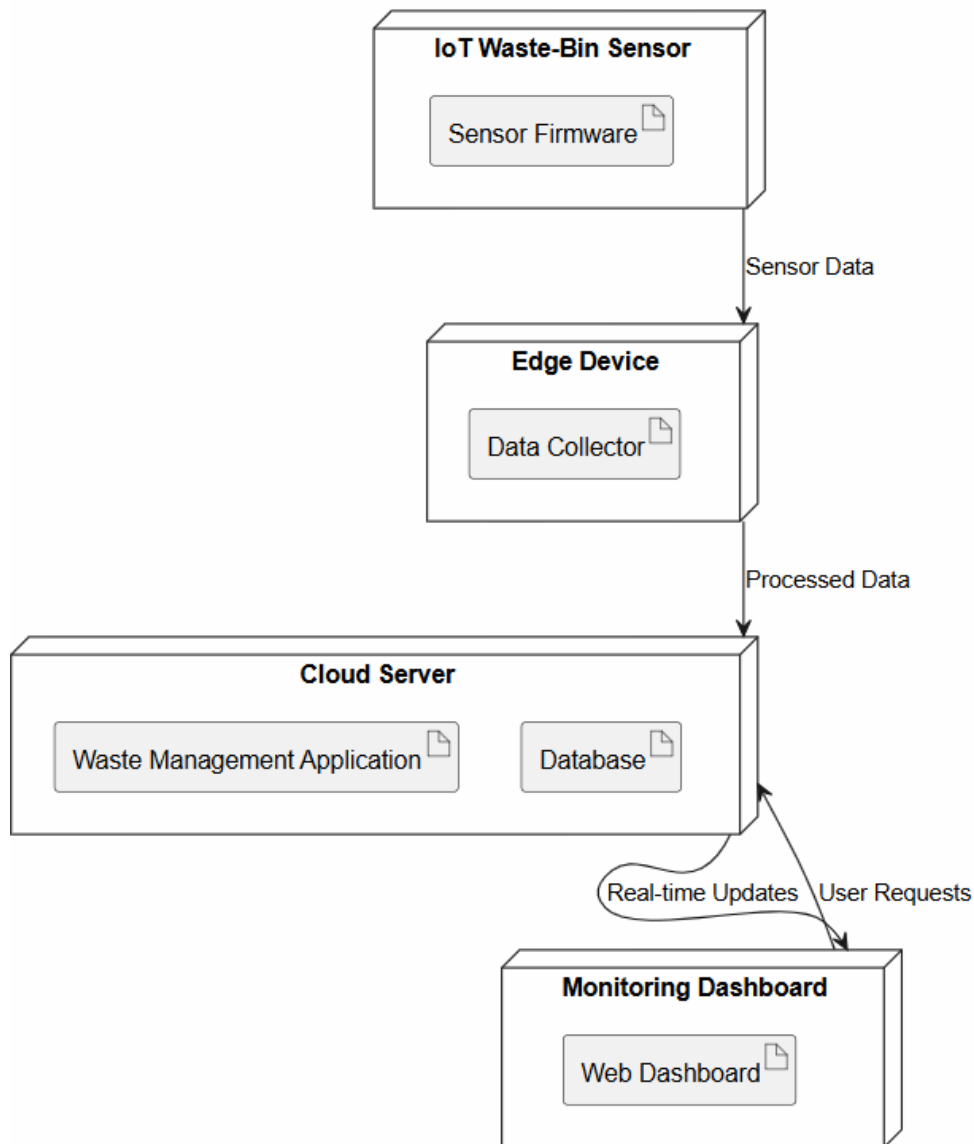
These interfaces enable communication between software components while maintaining modularity.

d) Conclusion

The Component Diagram represents the architecture of the Smart Healthcare Portal by identifying major software components, their interfaces, and dependencies. It provides a modular and scalable design that supports efficient interaction among healthcare services.

6. Deployment Diagram Design

A **Smart City Waste Management System** is deployed across IoT waste-bin sensors, edge devices, cloud servers, and monitoring dashboards. The current design lacks clarity regarding physical architecture and communication links.



System Analysis :

The Smart City Waste Management System uses IoT technology to monitor waste-bin levels in real time. Waste-bin sensors collect data and send it to edge devices for preliminary processing. The processed data is transmitted to cloud servers, where it is stored and analyzed. Monitoring dashboards allow city administrators to view waste status and manage collection activities. The Deployment Diagram illustrates the physical architecture, deployed software artifacts, and communication links between system nodes.

a) Deployment Nodes

- IoT Waste-Bin Sensor
- Edge Device
- Cloud Server
- Monitoring Dashboard

b) Artifacts

- Sensor Firmware – Collects waste-bin data.
- Data Collector – Processes and forwards sensor data.
- Waste Management Application – Analyzes and manages waste information.
- Database – Stores sensor readings and operational data.
- Web Dashboard – Displays waste-bin status and reports to administrators.

c) Communication Links

- IoT Waste-Bin Sensor → Edge Device: Transfers waste-bin sensor data.
- Edge Device → Cloud Server: Sends processed data to the cloud.
- Cloud Server → Monitoring Dashboard: Provides real-time updates and reports.
- Monitoring Dashboard → Cloud Server: Sends user requests and retrieves information.

These communication links ensure continuous monitoring and efficient waste management.

d) Conclusion

The Deployment Diagram represents the physical architecture of the Smart City Waste Management System by identifying deployment nodes, software artifacts, and communication links. It provides a clear understanding of how IoT devices, edge computing, cloud services, and monitoring dashboards interact to support efficient waste management.